

Project 2

Project description

You will need to implement a trace-driven cache simulator, and use it to evaluate the performance of different cache architecture features. The project is described in detail in `sim.pdf`.

File description

In the attachment, you will find the following project files:

- `cache.h` - cache simulator definitions
- `cache.c` - cache simulator (put all your edits here)
- `main.h` - simulation driver definitions
- `main.c` - simulation driver
- Public tests
 - `public-block.trace` - test cache block sizes
 - `public-assoc.trace` - test cache associativity
 - `public-write.trace` - test cache write policy
 - `public-instr.trace` - test instruction cache
 - `spice10.trace` - 1st 10 accesses in `spice.trace`
 - `spice100.trace` - 1st 100 accesses in `spice.trace`
 - `spice1000.trace` - 1st 1000 accesses in `spice.trace`
- Expected outputs for public tests
 - `public-block1.out`
 - `public-assoc1.out`
 - `public-write1.out`
 - `public-instr1.out`
 - `spice10.out`
 - `spice100.out`
 - `spice1000.out`
- `Makefile` - makefile to create simulator
- `runPublic` - `cs`h script to run public tests
- `sim.pdf` - detailed description of project
- `tags.txt` - example index & tag values for `spice100.trace`

In addition, there are three large application traces:

- `spice.trace` - circuit simulator
- `cc.trace` - C compiler
- `tex.trace` - Tex document processor

Requirements

- Write a simple report about how you implement the cache simulator and answer the questions in sim.pdf
- Compress your project and report, submit it to [XueZaiZheDa](#) before the deadline (20261.2). Use your student ID as filename. Files you should submit:
 - code/ directory
 - Report

Note

- It's ok if you can't implement a fully functional simulator, make your best effort and do what you can do.